General-r dim-additivity for $B^{(\lambda)}_{\ell+1}V_\lambda$ and the structural closed-form SYT formula

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Abstract

The May-13 morning paper on multi-corner dimensions conjectured a closed-form

$$\dim B^{(\lambda)}_{\ell+1}V_\lambda \;=\; f^{\lambda^{(\geq 2)}}$$

in the stable regime. The May-13 afternoon paper `2026-05-13-4-3-1n-lower-bound-closed.tex` proved an *r=2 dim-additivity meta-theorem* structurally for length-preserving rank $r = 2$, and Remark 9 of that paper sketched the r-general extension. The present paper formalises the r-general version and uses it to give a structural inductive proof of the closed-form SYT formula under a clean stability hypothesis.

Main theorem (general-r dim-additivity). Let λ be a partition with $r \geq 1$ length-preserving corners $c_1, \ldots, c_r$ and column-1 bottom corner c_*. Suppose the SRD-style reduction

$$B^{(\lambda)}_{\ell+1}V_\lambda \;=\; \mathcal{O}_\lambda\Big[\sum_{i=1}^{r}\Phi^\lambda_i\big(B^{(\mu_i)}V_{\mu_i}\big)\Big]$$

holds, where $\mu_i := \lambda \setminus \{c_i, c_*\}$, Φ^λ_i is the cell-pair branching injection, and $\mathcal{O}_\lambda$ is the outer chain. Suppose further that the corner-pair T_{n-1}-2-blocks $\{c_i, c_*\}$ are all non-degenerate (cells in distinct rows and columns; automatic). Then

$$\dim B^{(\lambda)}_{\ell+1}V_\lambda \;=\; \sum_{i=1}^{r}\dim B^{(\mu_i)}_{j_0(\mu_i)}V_{\mu_i}.$$

Corollary (closed-form SYT formula). If λ is in the *fully stable regime* — meaning that the SRD-style reduction above holds at every shape encountered in the recursive descent from λ to base case — then

$$\dim B^{(\lambda)}_{\ell+1}V_\lambda \;=\; f^{\lambda^{(\geq 2)}}.$$

The proof transcribes the r=2 argument verbatim: the unordered cell pairs $\{c_i, c_*\}$ are pairwise distinct (they share c_* but differ in the other element), so the seminormal SYT supports of the Φ^λ_i-images are pairwise disjoint; each image lies in $E^+_{n-1}|_{V_\lambda}$; and the outer chain $\mathcal{O}_\lambda$ acts injectively on $E^+_{n-1}|_{V_\lambda}$ by adjacent-injectivity. The corollary is a straightforward induction on $|\lambda|$ using the SYT branching rule.

We verify the theorem on r=3 shapes computationally.

1 Setup

Notation. As in the May-13 afternoon paper `2026-05-13-4-3-1n-lower-bound-closed.tex` (henceforth **LB-closed**). $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \ldots, T_{n-1}$ obeying $(T_i - q)(T_i + 1) = 0$ and the standard braid relations. V_λ is the seminormal cell module for

$\lambda \vdash n$ in asymmetric Murphy normalisation ($b' = 1$). $E_j^{\pm}$ denote the q- and (-1)-eigenspaces of T_j acting on V_λ (or any $H_q(S_n)$-module). For $1 \le k \le n$,

$$R'_k \;:=\; (T_{k-1}{+}1)\cdots(T_1{+}1), \qquad B^{(\lambda)}_{j_0} V_\lambda \;:=\; R'_{j_0}\cdots R'_n V_\lambda$$

with $j_0 := \ell + 1$, $\ell := \ell(\lambda)$. We always work at the sharp index $j_0 = \ell + 1$.

Partition data. Let $\lambda \vdash n$ have length $\ell = \ell(\lambda)$ and write λ as $(\hat\lambda, 1^q)$ with $\hat\lambda$ the part of λ with rows of length ≥ 2 and q the number of trailing 1-rows. The *length-preserving corners* of λ are the removable corners $(r, c) \in \lambda$ with $c \ge 2$; equivalently, the corners of $\hat\lambda$ that sit in column ≥ 2. List them as $c_1, \ldots, c_r$. The *column-*1 *bottom* is $c_* := (\ell, 1)$, which is a removable corner of λ exactly when $\lambda_\ell = 1$, i.e., $q \ge 1$. Throughout this paper we assume $q \ge 1$ so that c_* is a corner.

Decoration shape. $\lambda^{(\ge 2)} := (\lambda_1 - 1, \lambda_2 - 1, \ldots, \lambda_k - 1)$ where k is the number of rows of λ with $\lambda_i \ge 2$; equivalently, $\lambda^{(\ge 2)}$ is λ with column 1 deleted. There is an explicit bijection between length-preserving corners of λ and corners of $\lambda^{(\ge 2)}$: $(r, c) \mapsto (r, c - 1)$.

Corner-pair branching injections. For each length-preserving corner c_i, set

$$\mu_i \;:=\; \lambda \setminus \{c_i, c_*\}.$$

Note $|\mu_i| = n{-}2$, $\ell(\mu_i) = \ell{-}1$. The *branching injection* $\Phi_i^\lambda : V_{\mu_i} \hookrightarrow V_\lambda$ is defined on the seminormal basis by

$$\Phi_i^\lambda(v^{(\mu_i)}_{T'}) \;=\; v_{T_A^i} + \tau_i\, v_{T_B^i},$$

where $T_A^i, T_B^i \in \mathrm{SYT}(\lambda)$ are the two extensions of T' obtained by placing $\{n{-}1, n\}$ at $\{c_i, c_*\}$ in the two orders, and τ_i is the Hoefsmit off-diagonal coefficient of the T_{n-1}-2-block at the axial distance $d_i := \mathrm{ax}(c_i) - \mathrm{ax}(c_*)$ between the two cells (**LB-closed**, §1; cf. SRD-331 Lemma 3.2). Since c_i is in column ≥ 2 and a non-bottom row while c_* is in column 1 bottom row, $d_i \ne 0$ and the 2-block is non-degenerate. $\Phi_i^\lambda(v^{(\mu_i)}_{T'})$ is the $+q$-eigenvector of T_{n-1} on the 2-block $\langle v_{T_A^i}, v_{T_B^i}\rangle$.

Outer chain.

$$\mathcal{O}_\lambda \;:=\; (T_\ell{+}1)(T_{\ell+1}{+}1)\cdots(T_{n-2}{+}1),$$

an $(n - \ell - 1)$-factor product of $(T_j + 1)$'s with j running from ℓ up to $n - 2$.

SRD-style reduction (hypothesis). We *assume* for the purposes of Theorem 6 that

$$B^{(\lambda)}_{\ell+1} V_\lambda \;=\; \mathcal{O}_\lambda\Big[\,\sum_{i=1}^{r} \Phi_i^\lambda\big(B^{(\mu_i)} V_{\mu_i}\big)\Big]. \tag{1}$$

This is the form taken on by the SRD reduction when all "other" pieces vanish: same-row pieces (contributing the $+q$-eigenspaces of T_j with j inside row blocks of length ≥ 2) and pure-length-preserving pieces (contributing Φ_{ij}^λ at corner pairs $\{c_i, c_j\}$). Both vanishings have been proved in specific shape families:

- Same-row dies: proved for $(3, 3, 1^q)$ ($q \ge 2$) `2026-05-13-same-row-dies-331.tex`; for $(4, 2, 1^q)$ ($q \ge 2$) `2026-05-13-same-row-dies-421.tex`; and generally in the $r = 1$ inductive reduction `2026-05-13-r1-inductive-reduction.tex`.

- Pure-length-preserving dies: proved for $(4,3,1^q)$ via j-formula leakage on the $\{c_1,c_2\}$-block `2026-05-13-4-3-1n-upper-bound.tex` Lemma 3.5.

We treat (1) as a structural hypothesis on the shape λ. The present paper's content is the implication $((1)) \Rightarrow \dim B^{(\lambda)} = \sum_i \dim B^{(\mu_i)}$.

2 Disjoint cell-pair supports

The key combinatorial lemma generalises **LB-closed** Lemma 1 from $r=2$ to arbitrary r.

Lemma 1 (Disjoint SYT supports)**.** *For each $i \in \{1,\dots,r\}$ and each $v \in V_{\mu_i}$, every seminormal-basis SYT T in the support of $\Phi_i^\lambda(v) \in V_\lambda$ satisfies $\{T^{-1}(n-1), T^{-1}(n)\} = \{c_i, c_*\}$ as an unordered pair of cells. Consequently, for distinct $i \neq j$,*

$$\Phi_i^\lambda(V_{\mu_i}) \;\cap\; \Phi_j^\lambda(V_{\mu_j}) \;=\; \{0\}$$

inside V_λ, and the sum $\sum_{i=1}^r \Phi_i^\lambda(V_{\mu_i})$ is a direct sum.

Proof. By construction (§1), for a seminormal basis vector $v_{T'}^{(\mu_i)}$ of V_{μ_i},

$$\Phi_i^\lambda(v_{T'}^{(\mu_i)}) \;=\; v_{T_A^i} + \tau_i\, v_{T_B^i},$$

where T_A^i, T_B^i are the two SYTs of λ that restrict to T' on cells of μ_i and place $\{n-1,n\}$ at $\{c_i,c_*\}$ in the two orders. So both supporting SYTs have $\{T^{-1}(n-1),T^{-1}(n)\} = \{c_i,c_*\}$. Extending by linearity over $v = \sum_{T'} a_{T'} v_{T'}^{(\mu_i)} \in V_{\mu_i}$, every SYT in the support of $\Phi_i^\lambda(v)$ has $\{n-1,n\}$ at $\{c_i,c_*\}$.

For $i \neq j$, the unordered cell pairs $\{c_i,c_*\}$ and $\{c_j,c_*\}$ are distinct as subsets of cells of λ: they share c_* but differ in the other element ($c_i \neq c_j$, since $i \neq j$ and $c_1,\dots,c_r$ are distinct length-preserving corners). Hence no single SYT T of λ has its $\{T^{-1}(n-1),T^{-1}(n)\}$ pair equal to *both* $\{c_i,c_*\}$ and $\{c_j,c_*\}$ simultaneously; equivalently, the SYT supports of $\Phi_i^\lambda(V_{\mu_i})$ and $\Phi_j^\lambda(V_{\mu_j})$ are disjoint as subsets of $\mathrm{SYT}(\lambda)$.

The seminormal basis $\{v_T : T \in \mathrm{SYT}(\lambda)\}$ is linearly independent. So if $v_i \in \Phi_i^\lambda(V_{\mu_i})$ for $i=1,\dots,r$ and $\sum_{i=1}^r v_i = 0$, then expanding each v_i in the seminormal basis and using pairwise disjoint supports forces each $v_i = 0$ separately. Hence $\sum_i \Phi_i^\lambda(V_{\mu_i}) = \bigoplus_i \Phi_i^\lambda(V_{\mu_i})$. ☐

Lemma 2 (Φ_i^λ is injective)**.** *Each $\Phi_i^\lambda : V_{\mu_i} \to V_\lambda$ is injective.*

Proof. For distinct seminormal basis vectors $v_{T_1'}^{(\mu_i)}$ and $v_{T_2'}^{(\mu_i)}$ of V_{μ_i}, their images $\Phi_i^\lambda(v_{T_1'}^{(\mu_i)}) = v_{T_A^{i,1}} + \tau_i\, v_{T_B^{i,1}}$ and $\Phi_i^\lambda(v_{T_2'}^{(\mu_i)}) = v_{T_A^{i,2}} + \tau_i\, v_{T_B^{i,2}}$ have disjoint 2-element supports: $T_A^{i,1}, T_B^{i,1}$ are extensions of T_1' and $T_A^{i,2}, T_B^{i,2}$ are extensions of T_2'; if $T_A^{i,1} = T_A^{i,2}$ then restricting to cells of μ_i gives $T_1' = T_2'$, contradicting $T_1' \neq T_2'$.

Hence Φ_i^λ sends a basis of V_{μ_i} to vectors with pairwise disjoint, non-zero seminormal supports, which form a linearly independent set in V_λ. So Φ_i^λ is injective. ☐

Corollary 3 (Dimension of the inner direct sum)**.**

$$\dim\Big(\bigoplus_{i=1}^r \Phi_i^\lambda(B^{(\mu_i)} V_{\mu_i})\Big) \;=\; \sum_{i=1}^r \dim B^{(\mu_i)} V_{\mu_i}.$$

Proof. By Lemma 1, the sum is direct, so the dimension equals $\sum_i \dim \Phi_i^\lambda(B^{(\mu_i)})$. By Lemma 2, each Φ_i^λ is injective, hence preserves dimensions: $\dim \Phi_i^\lambda(B^{(\mu_i)}) = \dim B^{(\mu_i)}$. ☐

3 Image in E_{n-1}^+ and adjacent-injectivity

The next two ingredients are unchanged from **LB-closed**; we restate them for self-containment.

Lemma 4 (Φ_i^λ lands in E_{n-1}^+; **LB-closed** Lemma 2)**.** *For each $i \in \{1, \dots, r\}$ and every $v \in V_{\mu_i}$, $\Phi_i^\lambda(v) \in E_{n-1}^+|_{V_\lambda}$.*

Proof. By construction, $\Phi_i^\lambda(v_{T'}^{(\mu_i)})$ is the $+q$-eigenvector of T_{n-1} on the 2-block $\langle v_{T_A^i}, v_{T_B^i} \rangle$ of V_λ. The 2-block is non-degenerate because c_i and c_* lie in distinct rows and distinct columns (c_i is in column ≥ 2 non-bottom row; c_* is in column 1 bottom row). Hence the $+q$-eigenvector exists and is unique up to scalar, and $\Phi_i^\lambda(v_{T'}^{(\mu_i)}) \in E_{n-1}^+$. Extending by linearity, $\Phi_i^\lambda(V_{\mu_i}) \subseteq E_{n-1}^+$. □

Theorem 5 (Adjacent injectivity chain; **LB-closed** Theorem 4 / dim2-331 Theorem 3.4)**.** *For any $H_q(S_n)$-module V and $a \leq b \leq n-2$, the operator*

$$(T_a{+}1)(T_{a+1}{+}1)\cdots(T_b{+}1)\big|_{E_{b+1}^+(V)} \;:\; E_{b+1}^+(V) \;\longrightarrow\; E_a^+(V)$$

is injective. In particular, the outer chain $\mathcal{O}_\lambda = (T_\ell{+}1)\cdots(T_{n-2}{+}1)$ acts injectively on $E_{n-1}^+|_{V_\lambda}$, with image in $E_\ell^+|_{V_\lambda}$.

4 Main theorem: general-r dim-additivity

Theorem 6 (Main: general-r dim-additivity)**.** *Let $\lambda \vdash n$ have length ℓ, $r \geq 1$ length-preserving corners $c_1, \dots, c_r$, and column-1 bottom corner c_* (we assume $\lambda_\ell = 1$). Assume the SRD-style reduction (1) holds. Then*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda \;=\; \sum_{i=1}^{r} \dim B_{j_0(\mu_i)}^{(\mu_i)} V_{\mu_i} \tag{2}$$

where $\mu_i := \lambda \setminus \{c_i, c_*\}$ and $j_0(\mu_i) := \ell(\mu_i) + 1 = \ell$.

Proof. Define

$$W \;:=\; \sum_{i=1}^{r} \Phi_i^\lambda\big(B^{(\mu_i)} V_{\mu_i}\big) \;\subseteq\; V_\lambda.$$

We compute $\dim W$ and $\dim \mathcal{O}_\lambda W$.

Step 1. W is a direct sum of dimension $\sum_{i=1}^r \dim B^{(\mu_i)}$.

By Lemma 1 applied to the inclusion $B^{(\mu_i)} V_{\mu_i} \subseteq V_{\mu_i}$, the pieces $\Phi_i^\lambda(B^{(\mu_i)} V_{\mu_i})$ have pairwise disjoint seminormal-SYT supports inside V_λ. Hence they form a direct sum, and by Corollary 3, $\dim W = \sum_i \dim B^{(\mu_i)}$.

Step 2. $W \subseteq E_{n-1}^+|_{V_\lambda}$.

Each summand $\Phi_i^\lambda(B^{(\mu_i)} V_{\mu_i}) \subseteq \Phi_i^\lambda(V_{\mu_i}) \subseteq E_{n-1}^+|_{V_\lambda}$ by Lemma 4. The sum (direct or otherwise) of subspaces of $E_{n-1}^+|_{V_\lambda}$ remains in $E_{n-1}^+|_{V_\lambda}$.

Step 3. $\mathcal{O}_\lambda|_W$ is injective.

By Theorem 5, $\mathcal{O}_\lambda$ is injective on all of $E_{n-1}^+|_{V_\lambda}$, hence on the subspace W.

Step 4. Conclude.

By Step 3, $\dim \mathcal{O}_\lambda W = \dim W$. By the SRD-style reduction (1), $B^{(\lambda)} V_\lambda = \mathcal{O}_\lambda W$. So

$$\dim B^{(\lambda)} V_\lambda \;=\; \dim \mathcal{O}_\lambda W \;=\; \dim W \;=\; \sum_{i=1}^{r} \dim B^{(\mu_i)} V_{\mu_i}.$$

□

Remark 7 (Reading the proof)**.** The proof is a strict generalisation of **LB-closed** Theorem 6 (the $r = 2$ dim-additivity): replace "2+3" by "$\sum_i \dim B^{(\mu_i)}$" and "two corner pairs $\{c_1, c_*\}, \{c_2, c_*\}$" by "$r$ corner pairs $\{c_1, c_*\}, \dots, \{c_r, c_*\}$" throughout. The combinatorial heart (disjoint unordered cell pairs) generalises trivially because, while the cell pairs share c_*, they differ in the other element, and the other elements $c_1, \dots, c_r$ are pairwise distinct.

5 Closed-form SYT corollary by induction

Definition 8 (Fully stable regime)**.** A partition λ is in the *fully stable regime* if the SRD-style reduction (1) holds at λ and at every shape λ' encountered in the recursive descent

$$\lambda \longrightarrow \{\mu_i : 1 \le i \le r(\lambda)\} \longrightarrow \{\mu_i' : \dots\} \longrightarrow \cdots \longrightarrow (\text{column shape } (1^k)).$$

Equivalently, at every step, the same-row pieces and the pure length-preserving Φ_{ij}-pieces vanish.

Lemma 9 (Decoration shape under corner-pair removal)**.** *For λ with $\lambda_\ell = 1$ and length-preserving corner $c_i = (\rho_i, \kappa_i)$ ($\kappa_i \ge 2$), define $\mu_i := \lambda \setminus \{c_i, c_*\}$. Then*

$$\mu_i^{(\ge 2)} \;=\; \lambda^{(\ge 2)} \setminus c_i^{(\ge 2)},$$

where $c_i^{(\ge 2)} := (\rho_i, \kappa_i - 1)$ is the cell of $\lambda^{(\ge 2)}$ corresponding to c_i. Moreover, $c_i^{(\ge 2)}$ is a removable corner of $\lambda^{(\ge 2)}$, and the map $c_i \mapsto c_i^{(\ge 2)}$ is a bijection between length-preserving corners of λ and removable corners of $\lambda^{(\ge 2)}$.

Proof. First, the bijection between length-preserving corners of λ and corners of $\lambda^{(\ge 2)}$. A length-preserving corner $c_i = (\rho_i, \kappa_i) \in \lambda$ has $\kappa_i \ge 2$ and is removable, i.e., $\lambda_{\rho_i} = \kappa_i$ and $\lambda_{\rho_i+1} < \kappa_i$. Then $(\rho_i, \kappa_i - 1)$ is a cell of $\lambda^{(\ge 2)}$ (since $\kappa_i - 1 \ge 1$), and is removable there: $(\lambda^{(\ge 2)})_{\rho_i} = \lambda_{\rho_i} - 1 = \kappa_i - 1$, and $(\lambda^{(\ge 2)})_{\rho_i+1} = \max(\lambda_{\rho_i+1} - 1, 0) < \kappa_i - 1$.

Conversely, a removable corner $(\rho, \kappa) \in \lambda^{(\ge 2)}$ has $(\lambda^{(\ge 2)})_\rho = \kappa$, $(\lambda^{(\ge 2)})_{\rho+1} < \kappa$. Then $\lambda_\rho = \kappa + 1 \ge 2$, $\lambda_{\rho+1} \le \kappa$. So $(\rho, \kappa+1) \in \lambda$ is a removable corner in column ≥ 2, i.e., length-preserving.

Now compute $\mu_i^{(\ge 2)}$. μ_i is λ with c_i and c_* removed:

$$(\mu_i)_j = \begin{cases} \lambda_j - 1 & j = \rho_i, \\ \lambda_j & j \ne \rho_i, j \ne \ell, \\ \lambda_\ell - 1 = 0 & j = \ell. \end{cases}$$

So μ_i has $\ell - 1$ rows (the last row vanishes), and row ρ_i has $\kappa_i - 1$ cells.

The decoration shape $\mu_i^{(\ge 2)}$ removes column 1 from μ_i: row j of $\mu_i^{(\ge 2)}$ has $(\mu_i)_j - 1$ cells when $(\mu_i)_j \ge 2$; rows with $(\mu_i)_j \le 1$ become empty.

For $j = \rho_i$: $(\mu_i^{(\ge 2)})_{\rho_i} = \kappa_i - 2$ (if $\kappa_i \ge 3$) or 0 (if $\kappa_i = 2$, leaving an empty row). For $j \ne \rho_i, \ell$: $(\mu_i^{(\ge 2)})_j = \lambda_j - 1 = (\lambda^{(\ge 2)})_j$. For $j = \ell$: $(\mu_i^{(\ge 2)})_\ell = 0$ trivially.

On the other hand, $\lambda^{(\ge 2)} \setminus c_i^{(\ge 2)}$ removes the cell $(\rho_i, \kappa_i - 1)$ from $\lambda^{(\ge 2)}$. Row ρ_i becomes $\kappa_i - 2$ cells (if $\kappa_i \ge 3$) or becomes empty (if $\kappa_i = 2$). Other rows unchanged.

So $\mu_i^{(\ge 2)} = \lambda^{(\ge 2)} \setminus c_i^{(\ge 2)}$ matches row-by-row. □

Corollary 10 (Closed-form SYT formula)**.** *If λ is in the fully stable regime (Definition 8), then*

$$\dim B_{\ell+1}^{(\lambda)} V_\lambda \;=\; f^{\lambda^{(\ge 2)}}.$$

Proof. By strong induction on $n = |\lambda|$.

Base case. If λ has no length-preserving corners ($r = 0$), then λ is a single column (1^n) and $\lambda^{(\geq 2)} = \emptyset$. Here $j_0 = \ell + 1 = n + 1 > n$, so $B^{(\lambda)} = R'_{n+1} R'_{n+2} \cdots R'_n$ is the *empty product* (the index range is empty), equal to the identity. Hence $\dim B^{(\lambda)} V_\lambda = \dim V_{(1^n)} = 1 = f^{\emptyset}$. *Note.* In this case the SRD-style reduction is vacuous: there are no μ_i's.

Inductive step. Suppose λ has $r \geq 1$ length-pres corners. By Theorem 6 (applicable since the fully-stable hypothesis gives (1)),

$$\dim B^{(\lambda)} V_\lambda \;=\; \sum_{i=1}^{r} \dim B^{(\mu_i)} V_{\mu_i}.$$

Each μ_i has $|\mu_i| = n - 2 < n$, and the fully-stable regime of λ implies the fully-stable regime of each μ_i (the descent tree of μ_i is a subtree of the descent tree of λ). By induction hypothesis, $\dim B^{(\mu_i)} V_{\mu_i} = f^{\mu_i^{(\geq 2)}}$.

By Lemma 9, $\mu_i^{(\geq 2)} = \lambda^{(\geq 2)} \setminus c_i^{(\geq 2)}$, where $c_i^{(\geq 2)}$ ranges over all removable corners of $\lambda^{(\geq 2)}$ as i ranges over $\{1, \ldots, r\}$. By the SYT branching identity,

$$\sum_{c \text{ corner of } \lambda^{(\geq 2)}} f^{\lambda^{(\geq 2)} \setminus c} \;=\; f^{\lambda^{(\geq 2)}}.$$

Combining,

$$\dim B^{(\lambda)} V_\lambda \;=\; \sum_{i=1}^{r} f^{\mu_i^{(\geq 2)}} \;=\; \sum_{c \text{ corner of } \lambda^{(\geq 2)}} f^{\lambda^{(\geq 2)} \setminus c} \;=\; f^{\lambda^{(\geq 2)}}.$$

□

Remark 11 (On the fully-stable hypothesis)**.** The fully-stable regime hypothesis (Def. 8) is the substantive condition under which the closed-form formula holds. Concretely, it requires:

- At every shape λ' in the descent tree, the same-row pieces vanish.
- At every shape λ', the pure length-pres $\Phi_{ij}^{\lambda'}$ pieces (corresponding to corner pairs $\{c_i, c_j\}$ with c_* *not* in the pair) vanish.

These vanishings have been proved for various individual shape families (cf. the listed papers in §1). A general structural proof of stability has not yet been given; the shape-dependent threshold $q_0(\widehat{\lambda})$ documented in the May-13 `2026-05-13-decoration-iso-failed.tex` paper indicates that stability is genuinely shape-sensitive at small q. The corollary therefore reads: *whenever stability holds, the SYT formula follows.*

6 Computational verification on r=3 shapes

We verify Theorem 6 computationally on $r = 3$ shapes via the dim-recursion

$$\dim B^{(\lambda)} \;=\; \dim B^{(\mu_1)} + \dim B^{(\mu_2)} + \dim B^{(\mu_3)}.$$

Setup. Mod-P arithmetic at $P = 100\,003$, $q = 11$ (matches `compute_dim_B.py`).

Test family $(4,3,2,1^q)$**.** $\hat\lambda = (4,3,2)$, $\hat\lambda^{(\geq 2)} = (3,2,1)$, $f^{(3,2,1)} = 16$. Three length-preserving corners: $c_1 = (1,4)$, $c_2 = (2,3)$, $c_3 = (3,2)$. Inner shapes and inner decoration shapes:

$$\begin{aligned}
\mu_1 &= (3,3,2,1^{q-1}), & \mu_1^{(\geq 2)} &= (2,2,1), & f^{(2,2,1)} &= 5,\\
\mu_2 &= (4,2,2,1^{q-1}), & \mu_2^{(\geq 2)} &= (3,1,1), & f^{(3,1,1)} &= 6,\\
\mu_3 &= (4,3,1^{q}), & \mu_3^{(\geq 2)} &= (3,2), & f^{(3,2)} &= 5.
\end{aligned}$$

Predicted by Theorem 6: $\dim B^{(\lambda)} = 5+6+5 = 16$, matching $f^{(3,2,1)}$ via SYT branching.

Computational results. Mod-P rank at $P = 100\,003$, $q = 11$ (computational q, not to be confused with the trailing 1-row count, which we now call t for clarity in the table):

t	n	$\dim V_\lambda$	$\dim B^{(\lambda)}$	target 16	verdict
0	9	168	16	16	*matches*
1	10	768	24	16	non-stable
2	11	2310	40	16	non-stable
3	12	5632	**16**	16	✓ stable

Verification of additivity at $t = 3$**.** Computing each inner $\dim B^{(\mu_i)}$ separately at $t = 3$ (so inner shapes have trailing 1-row count $t-1 = 2$ for μ_1, μ_2 and $t = 3$ for μ_3):

shape	n	$\dim B$	target $(f^{\mu^{(\geq 2)}})$
$\mu_1 = (3,3,2,1^2)$	10	5	$f^{(2,2,1)} = 5$ ✓
$\mu_2 = (4,2,2,1^2)$	10	6	$f^{(3,1,1)} = 6$ ✓
$\mu_3 = (4,3,1^3)$	10	5	$f^{(3,2)} = 5$ ✓
sum		16	matches $\dim B^{(\lambda)} = 16$ ✓

The stability threshold for this family is $t \geq 3$, sharper than the crude $t \geq |\lambda^{(\geq 2)}| = 6$ bound.

Curious observation at $t = 0$**.** At $t = 0$ (the no-trailing-1-rows case $\lambda = \hat\lambda = (4,3,2)$, $n = 9$), the computed $\dim B = 16$ *matches* the closed-form target, even though our framework formally requires $t \geq 1$ for the column-1 corner c_* to exist. The matching-failing-matching pattern $\{0 : \checkmark, 1 : \times, 2 : \times, 3 : \checkmark, \ldots\}$ echoes the non-monotone-in-t behaviour observed for $\widehat{\lambda} = (4,2)$ in `2026-05-13-decoration-iso-failed.tex`. The $t = 0$ match suggests an independent reduction (with a different cell structure) operates at the no-trailing-rows boundary. We leave this for future work.

Scripts: `~/projects/scratch/2026-05-13-r-additivity/test_r3_fast.py`, `compute_dim_B_fast.py`.

7 Honest status

What is proved structurally. Theorem 6 (general-r dim-additivity) is proved structurally, modulo only the SRD-style reduction hypothesis (1). The proof transcribes the r=2 proof of **LB-closed** verbatim, with the combinatorial heart (disjoint cell pairs) generalising trivially.

What is conditional. Corollary 10 (closed-form SYT formula) requires the fully-stable regime hypothesis at all descent shapes. Stability is shape-dependent and is itself a substantial structural fact; it has been proved in individual shape families but not yet in full generality.

What this resolves. The Remark 9 of **LB-closed** sketched the general-r extension; the present paper formalises it. The closed-form SYT formula, which was conjectural in `2026-05-13-multi-corner-dimensior` becomes a *structural* theorem (under the fully-stable hypothesis) rather than a numerical coincidence.

Comparison to the failed decoration iso. The naive Hecke-iso upgrade was refuted in `2026-05-13-decoratic` there is no parabolic embedding of $H_q(S_{n-\ell})$ inside $H_q(S_n)$ acting on $B^{(\lambda)}V_\lambda$ as $V_{\lambda(\geq 2)}$. The present paper achieves the next best thing: a *numerical* equality of dimensions, derived structurally via direct decomposition of V_λ, without claiming any rep-theoretic content beyond counting.

Next steps.

- Prove the SRD-style reduction (1) for all stable shapes structurally (currently only proved in special families).
- Characterise the shape-dependent stability threshold $q_0(\widehat{\lambda})$ (the failed-iso paper's open problem).
- Determine whether $B^{(\lambda)}V_\lambda$ carries a Hecke action via exotic operators (Jucys–Murphy, intertwiners) that upgrade the numerical iso to a rep-theoretic one.